COURSE WELCOME PACKET & SYLLABUS

AGENTIC CODING

From first terminal command to shipping an AI-assisted, full-stack application.

HYBRID & SELF-PACED IN D2L BRIGHTSPACE · LIVE BLOCKS MON + WED, 6:00–8:00 PM

Welcome to Agentic Coding! Over the next 18 weeks you'll go from "I've never touched a terminal" to building and shipping a real web application with an AI coding assistant working alongside you. You'll master VS Code, the Linux command line, Docker, FastAPI, and Git/GitHub — then put those skills to work with Continue.dev running against Temple College's own AI server. No cloud fees, no GPU required: everything runs on your laptop plus the college's shared infrastructure.

The course is **self-paced**. Every lesson opens with a short concept video and ends with a hands-on lab you complete on your own machine; submitting the lab unlocks the next lesson. Twice a week we meet live — **Monday** I teach the week's material, **Wednesday** is a lab clinic where we fix whatever is blocking you. Come to both whenever you can, and move at your own speed either way.

Two things to do before your first live block: complete the 5-minute survey below, and confirm your laptop meets the requirements on page 2.

See you in class — Tim Carrender

Instructor · Temple College Workforce Development, in collaboration with ChiefOS

DO THIS FIRST · 5 MINUTES

Pre-Course Survey

tktechventure.com/agentic-coding

Tells us what machine and operating system you're bringing and where you're starting from, so day one runs smoothly for everyone. Scan the code or type the link into any browser.

18	3	7	70	2
WEEKS	PHASES	MODULES	HANDS-ON LABS	LIVE BLOCKS / WEEK

HOW THE COURSE WORKS

1
WATCH
Every lesson opens with a 2–3 minute concept video that sets up what you're about to do.

2
DO
Every lesson has a hands-on lab — real commands on your own machine, building real things.

3
SUBMIT
Each lab produces an artifact — a screenshot, command output, or repo link — submitted in D2L. Completing it unlocks the next lesson.

4
GET UNSTUCK LIVE
Two live blocks per week support you. Labs — not attendance — gate your progress, so you can move ahead or catch up anytime.

LIVE BLOCK	WHAT HAPPENS
Monday · 6:00–8:00 PM Instruction block	Recap of where the suggested pace puts everyone, live teaching of the week's concepts with demos, then we kick off the week's first lab together in the room.
Wednesday · 6:00–8:00 PM Lab clinic	Open working session. Bring whatever is blocking you — we debug it together so you leave with your labs moving again.

WHAT YOUR COMPUTER NEEDS

Everyone: a 64-bit laptop with **8 GB RAM** (16 GB recommended), **30 GB free disk space**, reliable internet, and **administrator rights** to install software. A free **GitHub account** is needed by week 10 (github.com — you can create it now). **No GPU required** — the AI models run on the college's shared server (2× RTX A6000); your laptop just connects to it.

WINDOWS
WINDOWS 10 / 11 + WSL2

- **Windows 10** (version 2004, build 19041, or later) or any **Windows 11**.
- **WSL2 with Ubuntu** — we install it together in Module 00. Expect one reboot; installing ahead of week 1 saves class time.
- Hardware **virtualization enabled** in BIOS/UEFI (usually already on; we'll help if Windows prompts you).
- Docker and all Linux labs run inside WSL2 — your files and terminal live there for the whole course.
- **Alternative:** if WSL2 won't work on your machine, you can do the entire course over SSH into the college's Linux server (set up in Module 00).

MAC
MACOS — APPLE SILICON OR INTEL

- **macOS 12 Monterey or later** — both Apple Silicon (M1/M2/M3/M4) and Intel Macs are fine.
- Uses the **built-in Terminal** — no WSL or extra setup needed to get a Unix shell.
- VS Code, Docker Desktop, and all developer tools install natively on macOS.
- For the Ubuntu-specific drills (`apt`, services), Mac students connect to the college's Linux server over **SSH** (set up in Module 00).

Not supported: Chromebooks and tablets (including iPads). This course lives in a real Linux/Unix shell throughout — you need a laptop you can install software on.

SYLLABUS — THE 18-WEEK PACE

This schedule is a **suggested pace**, not a lock: it tells you roughly where to be each week and sets the focus of that week's live blocks. Because the course is self-paced, you may move faster or slower — each module ends in a **final checkpoint** (highlighted rows) that must be passed to advance to the next.

PHASE 1 FOUNDATIONS

Weeks 1–6 · Modules 00–02 · 27 labs

WK	MODULE	FOCUS
1	00 · VS Code	What an IDE is, installing VS Code, UI tour, the integrated terminal, extensions
2	00 → 01	Remote connections (WSL / SSH), **Module 00 checkpoint**; terminal setup + navigation
3	01 · Terminal	Creating files & folders, reading files, sudo & updates, installing software, permissions
4	01 · Terminal	Networking, processes & services, pipes & grep, users & groups, **Module 01 checkpoint**
5	02 · Docker	Docker install, your first container (hello-world), running a real image (nginx)
6	02 · Docker	Images vs containers, the VS Code Docker sidebar, **Module 02 checkpoint** + buffer

PHASE 2 BUILDING REAL APPS

Weeks 7–12 · Modules 03–04 · 25 labs

WK	MODULE	FOCUS
7	03 · FastAPI	Python virtual environments, installing FastAPI, hello world, path & query parameters
8	03 · FastAPI	Pydantic request bodies, routers, dependency injection, containerizing your API
9	03 · FastAPI	Docker Compose + Postgres, SQLAlchemy database, error handling, **Module 03 checkpoint**
10	04 · Git & GitHub	Configuring git, your first commits, the status/add/commit cycle, diff & log
11	04 · Git & GitHub	Branches, undoing changes, GitHub + SSH keys, pushing, cloning & pulling
12	04 · Git & GitHub	Pull requests — open, review a classmate's, merge — **Module 04 checkpoint** + buffer

PHASE 3 AI-ASSISTED DEVELOPMENT

Weeks 13–18 · Modules 05–06 + capstone · 18 labs

WK	MODULE	FOCUS
13	05 · Continue.dev	Meet the college's shared AI endpoint, connect to it, explore the models, verify chat
14	05 · Continue.dev	Configure Continue.dev in VS Code, chat with @-context, tab autocomplete
15	05 · Continue.dev	Inline edit, agent mode with tool-call approvals, **Module 05 checkpoint**
16	06 · HeroUI	Node.js via nvm, scaffolding a Next.js + HeroUI app, your first components
17	06 · HeroUI	CORS + fetching your FastAPI data, create-item form, extend with AI, **Module 06 checkpoint**
18	Capstone	Combine FastAPI + HeroUI + Continue.dev into a finished feature; **demo at the final session**

WHAT YOU'LL BUILD, MODULE BY MODULE

MODULE 00 · 8 LESSONS

VS Code & IDE Fundamentals

Your editor for the whole course: UI, command palette, integrated terminal, extensions, and remote connections over WSL and SSH.

MODULE 01 · 12 LESSONS

Ubuntu Terminal Fundamentals

Navigate, create, and inspect files; sudo & package installs; permissions; networking; processes; pipes & grep; users & groups.

MODULE 02 · 7 LESSONS

Docker Fundamentals

Install Docker, run your first containers, understand images vs containers, and manage everything from the VS Code sidebar.

MODULE 03 · 13 LESSONS

FastAPI Development

Build a real Python API: routes, validation with Pydantic, dependency injection, then containerize it with a Postgres database.

MODULE 04 · 12 LESSONS

Git & GitHub

Version control from first commit to branches, pushing to GitHub over SSH, and reviewing classmates' pull requests.

MODULE 05 · 10 LESSONS

AI Coding with Continue.dev

Connect VS Code to the college's AI server: chat with codebase context, autocomplete, inline edits, and agent mode.

MODULE 06 + CAPSTONE · 8 LESSONS

HeroUI Frontend & Final Project

Put a modern React/Next.js interface on top of your own API — built with AI assistance — then combine everything into a capstone feature you demo at the final session.

BEFORE YOUR FIRST LIVE BLOCK

- [ ] **Complete the pre-course survey** — **tktechventure.com/agentic-coding** (5 minutes; helps us prep for your exact machine).
- [ ] **Check your laptop** against the Windows / Mac requirements on page 2.
- [ ] **Windows students:** if you're comfortable, install WSL2 + Ubuntu ahead of time (it needs a reboot) — otherwise we'll do it together in week 1.
- [ ] **Create a free GitHub account** at github.com — not needed until week 10, but easiest to do now.
- [ ] **Log in to D2L Brightspace** and confirm you can see the course.

TIM CARRENDER

INSTRUCTOR · TEMPLE COLLEGE WORKFORCE DEVELOPMENT × CHIEFOS

Email: tim.carrender@gmail.com · **Live blocks:** Monday + Wednesday, 6:00–8:00 PM

Questions before the course starts? Email anytime — include "Agentic Coding" in the subject.

PRE-COURSE SURVEY

tktechventure.com/agentic-coding